Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial #4 – sterilization - Solutions

Question 1

A factory produces protein which is secreted by a certain yeast, in a production fermenter with the volume of 1 m^3. The carbon source in the fermentation is provided by a molasses (65% solids) with initial concentration of 30%, and the initial contamination of the molasses is 10^4 bacteria per gram. Additionally, the medium contains 15 gr/lit salts with an initial contamination of 10^6 bacteria per kilogram, and water which completes the fermenter volume and contains initial contamination of $2 \cdot 10^3$ bacteria per milliliter.

The medium is sterilized at a temperature of 132°C (K_d=1.75 min^{-1}), then it is being inoculated by 400 gr yeast and the fermentation immediately starts with logarithmic growth and ends when the yeast concentration in the fermenter reaches to 2.2%. The fermentation requirements enable bacterial contamination of up to 1,000 cells per fermenter at the end of the fermentation.

Additional data: the yeast growth rate is 0.267 hr^{-1} and the bacteria growth rate is 0.14 per hour.

What is the minimal sterilization period required? *(10 min)*

Solution:

Given:

Sterilization	**Contaminating bacteria**	**Yeast**	**Fermenter**
$T = 132°C = 405°K$	$N'_{0_{\text{molasses}}} = 10^4 \dfrac{cells}{g\text{ molasses}}$ $N'_{0_{\text{salt}}} = 10^6 \dfrac{cells}{Kg\text{ salt}}$ $N'_{0_{H_2O}} = 2 \cdot 10^3 \dfrac{cells}{mL}$ $N_f = 1000 \dfrac{cells}{Ferm}$ $k_d = 1.75 \dfrac{1}{min}$ $\mu_{bacteria} = 0.14 \dfrac{1}{hr}$	$WX_0 = 400gX$ $X_{f_{\text{yeast}}} = 2.2\% = 22 \dfrac{g\text{X}}{L}$ $\mu_{\text{yeast}} = 0.267 \dfrac{1}{hr}$	$V_{ferm} = 1m^3 = 1000L$ $S_{0_{\text{molasses}}} = 30\%$ $= 300 \dfrac{g\text{ molasses}}{L}$ $0.65 \dfrac{g\text{ solids}}{g\text{ molasses}}$ $S_{0_{\text{salt}}} = 15 \dfrac{g\text{ salt}}{L}$

We need to find the minimal sterilization period, t_{st}, that can be calculated from the sterilization equation:

$$ln \frac{N_1}{N_0} = -k_d \times t_{st}$$

Where k_d is given in the question, N_0 and N_1 represent the number of contaminating bacteria before and after sterilization, respectively.
The number of contaminating bacteria at the beginning of the fermentation we can calculate from the logarithmic growth equation:

$$ln \frac{X_f}{X_0} = \mu \times \Delta t$$

From the yeast parameters we can find the growth period in the production fermenter.
X_f and μ are given in the question, and X_0 can be calculated from the number of yeast that were inoculated in the fermenter:

$$X_0 = \frac{WX_0}{V}$$

$$X_0 = \frac{400gX}{1000L} = 0.4 \frac{gX}{L}$$

Therefore, the growth period in the production fermenter:

$$\Delta t = \frac{ln\frac{X_f}{X_0}}{\mu_{yeast}}$$

$$\Delta t = \frac{ln\left(\frac{22\frac{gX}{L}}{0.4\frac{gX}{L}}\right)}{0.267\frac{1}{hr}} = 15hr$$

We can use the fermentation requirements referring to the number of bacteria at the end of the fermentation in order to find the number of bacteria at its beginning:

$$ln\frac{N_f}{N_1} = \mu_{bacteria} \times \Delta t$$

$$ln\frac{1000\frac{cells}{Ferm}}{N_1} = 0.14\frac{1}{hr} \times 15hr \rightarrow N_1 = 122.456\frac{cells}{Ferm}$$

We still need to find the amount of contaminating bacteria before sterilization. In practice, the total number of bacteria is the sum of the bacteria originating from the different components of the growth medium:

$$N_0 = N_{0_{molasses}} + N_{0_{\text{salt}}} + N_{0_{\text{H}_2O}}$$

We will find the number of bacteria originating from each component.

The number of bacteria come from the salt solution:

$$N_{0_{\text{salt}}} = N'_{0_{\text{salt}}} \times S_{0_{\text{salt}}} \times V_{ferm}$$

$$N_{0_{\text{salt}}} = \frac{10^6\frac{cells}{Kg\text{ salt}} \times 15\frac{g\text{ salt}}{L} \times 1000\frac{L}{Ferm}}{10^3\frac{g\text{ salt}}{Kg\text{ salt}}} = 1.5 \times 10^7\frac{cells}{Ferm}$$

The number of bacteria come from the molasses:

$$N_{0_{\text{molasses}}} = N'_{0_{\text{molasses}}} \times S_{0_{\text{molasses}}} \times V_{ferm}$$

$$N_{0_{\text{molasses}}} = 10^4\frac{cells}{g\text{ molasses}} \times 300\frac{g\text{ molasses}}{L} \times 1000\frac{L}{Ferm} = 3 \times 10^9\frac{cells}{Ferm}$$

To find the number of bacteria come from the <u>water</u>, we need to first find the volume of water used to fill up the fermenter. The fermenter volume consists of the molasses volume and the water volume, so:

$$V_{\mathrm{H_2}O} = V_{Ferm} - V_{\mathrm{molasses}}$$

Given that the molasses contains 65% solids, meaning 35% water, therefore, the molasses volume added to the fermenter is:

$$V_{\mathrm{molasses}} = S_{0_{\mathrm{molasses}}} \times V_{ferm} \times \frac{g \text{ water}}{g \text{ molasses}}$$

$$V_{\mathrm{molasses}} = \frac{300 \frac{g \text{ molasses}}{L_{Ferm}} \times 1000 \frac{L_{Ferm}}{Ferm} \times 0.35 \frac{g \text{ water}}{g \text{ molasses}}}{1000 \frac{g \text{ water}}{L}} = 105 \frac{L}{Ferm}$$

Therefore, the water volume used to fill up the fermenter is:

$$V_{\mathrm{H_2}O} = (1000 - 105) \frac{L}{Ferm} = 895 \frac{L}{Ferm}$$

Hence, the number of bacteria come from the <u>water</u> is:

$$N_{0_{\mathrm{H_2}O}} = N'_{0_{\mathrm{H_2}O}} \times V_{\mathrm{H_2}O}$$

$$N_{0_{\mathrm{H_2}O}} = 2 \times 10^3 \frac{cells}{mL} \times 10^3 \frac{mL}{L} \times 895 \frac{L}{Ferm} = 1.79 \times 10^9 \frac{cells}{Ferm}$$

Therefore, the number of bacteria in the <u>fermenter</u>, before its sterilization, is:

$$N_0 = N_{0_{molasses}} + N_{0_{\mathrm{salt}}} + N_{0_{\mathrm{H_2}O}}$$

$$N_0 = 3 \times 10^9 \frac{cells}{Ferm} + 1.5 \times 10^7 \frac{cells}{Ferm} + 1.79 \times 10^9 \frac{cells}{Ferm} = 4.8 \times 10^9 \frac{cells}{Ferm}$$

Now we can go back to the sterilization equation in order to find the sterilization period:

$$ln \frac{N_1}{N_0} = -k_d \cdot t_{st}$$

$$ln\frac{122.456\frac{cells}{Ferm}}{4.8\times10^{9}\frac{cells}{Ferm}}=-1.75\frac{1}{min}\times t_{st}$$

$$\boxed{\boldsymbol{t_{st}=9.99\ min\cong10\ min}}$$

Question 2

A batch fermentation for growth of yeast cells is carried in a 1,500 liter fermenter. The medium contains molasses as the carbon source (69% solids while 58% of the solids is sugar), ammonia salts as the nitrogen source and Thiamine vitamin which is required in this fermentation. At the beginning of the fermentation the sugar concentration should be 40 gr/lit and the Thiamine concentration should be 200 ppm.

After medium preparation and before its inoculation the medium goes through a batch sterilization inside the fermenter. The sterilization is performed in a temperature of 125°C for 16 minutes (heating time of the medium to the sterilization temperature, as well as the cooling time can be neglected). The main source of contamination in the medium originates from the molasses, which contains spore concentration of 10^7 spores per gram molasses, and the sterilization process was planned so that the contamination probability will be only 1 to every 500 fermentations.

Additional data: the yeast growth yield is 0.674 gram yeast to gram molasses sugar, doubling time of yeast in this medium is 4455 seconds, the initial concentration of the yeast inoculated in the medium is 100 ppm, decomposition constant of Thiamine in the sterilization temperature is 4.12 hr^{-1}, the yeast start logarithmic growth immediately after inoculation.

A. What is the death constant of the spores in the sterilization temperature?

($2.14\ min^{-1}$)

B. Thiamine is provided to the medium as a concentrated stock solution with Thiamine concentration of 9%. What is the solution volume required to be added to the fermenter when preparing the medium? *(10 lit)*

C. Assuming the molasses sugar are the limiting factor in the medium – how long will it takes for the fermentation to reach stationary phase? *(10 hr)*

The sterilization process is performed automatically, but due to a computer error, 6 hours after the fermentation begun, the fermenter sterilization process started again. Luckily, the operation engineer in the factory passed across the fermenter, noticed the problem and manually stopped the sterilization process after 5 minutes. At that stage, the medium was immediately cooled down to the fermentation temperature, and after a 1 hour lag stage, the yeast started again the logarithmic growth.

D. Assuming the death coefficient of the yeast in the sterilization temperature is 0.35 per minute, what will be their concentration when they reach the stationary phase? *(17.62 grX/lit)*

E. How long the fermentation will last from its beginning to the stationary phase? *(~13.4 hr)*

F. What should be done to make sure that the yeast concentration at the end of this fermentation will be identical to their concentration at the end of a standard, normal fermentation? Detail all the assumptions and calculation required.

<u>Solution</u>:

Given:

<u>Yeast</u>	<u>Sterilization</u>	<u>Molasses</u>	<u>Fermenter</u>
$X_0 = 100ppm = 0.1\frac{gX}{L}$	$T = 125°\text{C}$	$0.69\frac{g_{\text{solid}}}{g_{molasses}}$	$V_{Ferm} = 1500L$
$t_d = 4455\,sec = 1.24\,hr$	$t_{st} = 16\,min$	$0.58\frac{g_{\text{sugar}}}{g_{solid}}$	$S_{0_{\text{sugar}}} = 40\frac{gS_{\text{sugar}}}{L}$
$Y_{\frac{X}{S}} = 0.674\frac{gX}{gS_{\text{sugar}}}$	$N_f = \frac{1}{500}\frac{contamination}{Ferm}$	$N_0' = 10^7\frac{Spore}{g_{molasses}}$	$C_1 = 200\,ppm = 0.2\frac{gT}{L}$
	$k' = 4.12\frac{1}{hr}$		

A. $\boldsymbol{k_d}$ =?

We can find k_d from the sterilization equation:

$$ln\frac{N_f}{N_0} = -k_d \times t_{st}$$

Given in the question - t_{st}.

We don't have any other data so we will assume that in this fermentation, contamination is caused by 1 spore which can help us find N_f:

$$N_f = \frac{1}{500} \frac{Contamination}{Ferm} = 0.02 \frac{Spore}{Ferm}$$

We will calculate N_0 by using N_0':

$$N_0 = \frac{N'_{0\text{molasses}} \times S_{0\text{sugar}} \times V_{ferm}}{\frac{g_{\text{solid}}}{g_{\text{molasses}}} \times \frac{g_{\text{sugar}}}{g_{solid}}}$$

$$N_0 = \frac{10^7 \frac{Spore}{g_{\text{molasses}}} \times 40 \frac{gS_{\text{sugar}}}{L} \times 1500 \frac{L}{Ferm}}{0.69 \frac{g_{\text{solid}}}{g_{\text{molasses}}} \times 0.58 \frac{g_{\text{sugar}}}{g_{solid}}} = 1.5 \times 10^{12} \frac{Spore}{Ferm}$$

From the sterilization equation we get that:

$$ln \frac{\frac{1}{500} \frac{Spore}{Ferm}}{1.5 \times 10^{12} \frac{Spore}{L}} = -k_d \times 16min$$

$$\boxed{\boldsymbol{k_d = 2.14 \frac{1}{min}}}$$

B. $\boldsymbol{V_{T\ stock}}$ =?

While it is given that $C_{\text{T stock}} = 9\% = 90 \frac{gT}{L}$.

The Thiamine concentration before sterilization we can find by using the decomposition equation:

$$ln \frac{C_1}{C_0} = -k' \times t_{st}$$

$$ln \frac{0.2 \frac{gT}{L}}{C_0} = -\frac{4.12 \frac{1}{hr}}{60 \frac{min}{hr}} \times 16\text{min} \rightarrow C_0 = 0.6 \frac{gT}{L}$$

Therefore, according to $C_1 \times V_1 = C_2 \times V_2$:

$$C_{\text{T stock}} \times V_{T\text{ stock}} = C_0 \times V_{ferm}$$

$$90 \frac{gT}{L} \times V_{T\text{ stock}} = 0.6 \frac{gT}{L} \times 1500L$$

$$\boxed{\boldsymbol{V_{T\text{ stock}} = 10L}}$$

C. $\Delta \boldsymbol{t} = ?$

The fermentation period we can find from the logarithmic growth equation:

$$ln\frac{X_f}{X_0} = \mu \cdot \Delta t$$

Given in the question X_0, so we need to find μ and X_f.

We can calculate μ from the doubling time of the yeast:

$$\mu = \frac{ln2}{t_d}$$

$$\mu = \frac{ln2}{1.24hr} = 0.56\frac{1}{hr}$$

We will calculate X_f from the growth yield on the molasses sugar, assuming the molasses sugar are the limiting factor in the medium. We will assume that the yeast enter the stationary stage when $S_f = 0\frac{gS_{\text{suagr}}}{L}$:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_f - X_0}{S_0 - S_f}$$

$$0.674\frac{gX}{gS_{\text{sugar}}} = \frac{(X_f - 0.1)\frac{gX}{L}}{(40 - 0)\frac{gS_{\text{sugar}}}{L}} \rightarrow X_f = 27.06\frac{gX}{L}$$

Therefore, we can find that the fermentation period is:

$$ln\frac{X_f}{X_0} = \mu \times \Delta t \rightarrow \Delta t = \frac{ln\frac{X_f}{X_0}}{\mu}$$

$$\Delta t = \frac{ln\left(\frac{27.06\frac{gX}{L}}{0.1\frac{gX}{L}}\right)}{0.56\frac{1}{hr}}$$

$$\boxed{\boldsymbol{\Delta t = 10\ hr}}$$

D. $\boldsymbol{X_f} = ?$

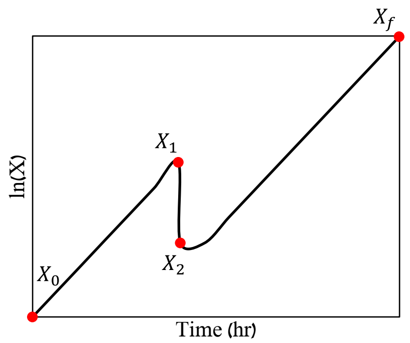
Given that:

$\Delta t_{log1} = 6hr$

$t_{st} = 5min$

$\Delta t_{lag} = 1hr$

$k_{dX} = 0.35\frac{1}{min}$



In the described malfunction scenario, 2 factors can lead to a decrease in the final biomass concentration: (I) the substrate, that was consumed before the malfunction, thus its concentration became lower, and (II) the Thiamin, that was also consumed like the substrate but was also damaged during the additional sterilization of the fermenter. Therefore, we need to find the biomass concentration that can be obtained at the point when these factors (the substrate and Thiamin) finish.

Solution rational:

1. The biomass concentration a moment before the malfunction is the initial concentration of the sterilization stage. We will calculate it from the growth equation.
2. The biomass concentration a moment after the malfunction is the initial concentration of the second growth stage. The sterilization death coefficient and the sterilization period are given, so from the sterilization equation we can find this concentration.
3. For the **substrate** as the limiting factor:
   a) By using the biomass concentration at the moment before the malfunction, we can find the remaining substrate concentration in the fermenter at that moment from the growth yield.
   b) By using the growth yield and the biomass concentration at the moment after the malfunction, we can calculate the final yeast biomass concentration that can be obtained.
4. For the **Thiamine** as the limiting factor:
   a) Thiamine is an expensive ingredient, so we will assume that in a normal fermentation it is completely consumed. We will calculate the biomass yield on the Thiamine from the normal fermentation process.
   b) From the biomass yield on the Thiamine and the biomass concentration a moment before the malfunction, we can calculate the Thiamine concentration remained in the medium at that moment.
   c) From the decomposition equation we can calculate the Thiamine concentration in the medium a moment after the malfunction.

d) From the biomass yield on Thiamine and the biomass concentration a moment after the malfunction, we will calculate the final biomass concentration of the yeast.

From the logarithmic growth equation, we will find the yeast concentration a moment before the malfunction:

$$ln\frac{X_1}{X_0} = \mu \times \Delta t_{log1}$$

$$ln\frac{X_1}{0.1\frac{gX}{L}} = 0.56\frac{1}{hr} \times 6hr \rightarrow X_1 = 2.88\frac{gX}{L}$$

From the sterilization equation we will find the yeast concentration a moment after the malfunction:

$$ln\frac{X_2}{X_1} = -k_{dX} \times t_{st}$$

$$ln\frac{X_2}{2.88\frac{gX}{L}} = -0.35\frac{1}{min} \times 5min \rightarrow X_2 = 0.5\frac{gX}{L}$$

From the growth yield we will find the sugar concentration remained in the medium before the malfunction:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_1 - X_0}{S_0 - S_1}$$

$$0.674\frac{gX}{gS_{\text{sugar}}} = \frac{(2.88 - 0.1)\frac{gX}{L}}{(40 - S_1)\frac{gS_{\text{sugar}}}{L}} \rightarrow S_1 = 35.88\frac{gS_{\text{sugar}}}{L}$$

This is the amount of sugar the yeast can grow on after the malfunction, so by the same yield we will find the yeast concentration when the molasses sugar is over:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_{fs} - X_2}{S_1 - S_f}$$

$$0.674\frac{gX}{gS_{\text{sugar}}} = \frac{\left(X_{fs} - 0.5\right)\frac{gX}{L}}{(35.88 - 0)\frac{gS_{\text{sugar}}}{L}}$$

$$\boldsymbol{X_{fs} = 24.68\frac{gX}{L}}$$

In the question it is given that Thiamine is vital for the fermentation, so we can assume it is completely consumed.

We will calculate the yeast yield on Thiamin in a normal fermentation process:

$$Y_{\frac{X}{C}} = \frac{\Delta X}{\Delta C} = \frac{X_f - X_0}{C_0 - C_f}$$

$$Y_{\frac{X}{C}} = \frac{(27.06 - 0.1)\frac{gX}{L}}{(0.2 - 0)\frac{gT}{L}} = 134.8\frac{gX}{gT}$$

From here, the Thiamine concentration in the medium a moment <u>before</u> the malfunction:

$$Y_{\frac{X}{C}} = \frac{X_1 - X_0}{C_0 - C_1}$$

$$134.8\frac{gX}{gT} = \frac{(2.88 - 0.1)\frac{gX}{L}}{(0.2 - C_1)\frac{gT}{L}} \rightarrow C_1 = 0.179\frac{gT}{L}$$

From the decomposition equation we will find the Thiamine concentration a moment <u>after</u> the malfunction:

$$ln\frac{C_2}{C_1} = -k' \times t_{st}$$

$$ln\frac{C_2}{0.179\frac{gT}{L}} = -\frac{4.12\frac{1}{hr}}{60\frac{min}{hr}} \times 5min \rightarrow C_2 = 0.127\frac{gT}{L}$$

This is the amount of Thiamine available for the yeast after the malfunction, so from the yeast yield on the Thiamine we can find the yeast concentration when the Thiamine in the medium is over:

$$Y_{\frac{X}{C}} = \frac{\Delta X}{\Delta C} = \frac{X_{fT} - X_2}{C_2 - C_f}$$

$$134.8\frac{gX}{gT} = \frac{\left(X_{fT} - 0.5\right)\frac{gX}{L}}{(0.127 - 0)\frac{gT}{L}}$$

$$\boxed{\boldsymbol{X_{f_T} = 17.62\frac{gX}{L}}}$$

After the malfunction, the Thiamine is the limiting factor. Therefore, the final concentration of the biomass in this fermentation is $\boldsymbol{17.62\frac{gX}{L}}$.

E. $\boldsymbol{\Delta t_T}$ =?

The total fermentation period includes the growth time before the malfunction, the lag time and the growth period after the malfunction:

$$\Delta t_T = \Delta t_{log1} + \Delta t_{lag} + \Delta t_{log2}$$

t_{log1} and Δt_{lag} are given in the question. From the logarithmic growth equation, we can find the growth period after the malfunction:

$$ln\frac{X_f}{X_2} = \mu \times \Delta t_{log2}$$

$$ln\frac{17.62\frac{gX}{L}}{0.5\frac{gX}{L}} = 0.56\frac{1}{hr} \times \Delta t_{log2} \rightarrow \Delta t_{log2} = 6.4hr$$

Therefore, the fermentation period in the malfunction scenario:

$$\Delta t_T = \Delta t_{log1} + \Delta t_{lag} + \Delta t_{log2}$$

$$\Delta t_T = (6 + 1 + 6.4)hr$$

$$\boxed{\boldsymbol{\Delta t_T = 13.4\ hr}}$$

F.

In section D we found that in order to retain the same final yeast concentration we need to add both molasses and Thiamine after the malfunction discovered.

Sugar is not ruined during sterilization, thus the amount of sugar that should be added can be calculated from the difference between the biomass concentration at the end of a <u>normal</u> process (as was calculated in section C) and the biomass concentration at the end of the <u>disrupted</u> process (as was calculated in section D according to the limiting substrate concentration):

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_{f\ normal} - X_{fS}}{\Delta S_{\text{sugar}}}$$

$$0.674\frac{gX}{gS_{\text{sugar}}} = \frac{(27.06 - 24.68)\frac{gX}{L}}{\Delta S_{\text{sugar}}} \rightarrow \Delta S_{\text{sugar}} = 3.53\frac{gS_{\text{sugar}}}{L}$$

We need to convert the unit from sugar to molasses:

$$W_{S_{\text{molasses}}} = \frac{\Delta S_{\text{sugar}} \times V_{Ferm}}{\frac{g_{\text{solid}}}{g_{molasses}} \times \frac{g_{\text{sugar}}}{g_{\text{solid}}}}$$

$$W_{S_{\text{molasses}}} = \frac{3.53 \frac{gS_{\text{sugar}}}{L} \times 1500L}{0.69 \frac{g_{\text{solid}}}{g_{molasses}} \times 0.58 \frac{g_{\text{sugar}}}{g_{\text{solid}}}} = \mathbf{13,230.1}\ \boldsymbol{g_{molasses}}$$

$$\boxed{\boldsymbol{W_{S_{\text{molasses}}} = 13.23\ Kg_{molasses}}}$$

Now we need to find the Thiamine volume that should be added to the medium after the malfunction. From the biomass yield on the Thiamine we can find the Thiamine concentration that should be added to the medium:

$$Y_{\frac{X}{C}} = \frac{\Delta X}{\Delta C} = \frac{X_{f\ \text{normal}} - X_{fT}}{\Delta C}$$

$$134.8 \frac{gX}{gT} = \frac{(27.06 - 17.62) \frac{gX}{L}}{\Delta C} \rightarrow \Delta C = 0.07 \frac{gT}{L}$$

From the fermenter volume we will calculate the amount of Thiamine that should be added:

$$W_T = \Delta C \times V_{Ferm}$$

$$W_T = 0.07 \frac{gT}{L} \times 1500L = \mathbf{105}\ \boldsymbol{gT}$$

The question gives us the Thiamine stock concentration, $C_{\text{T stock}} = 9\% = 90 \frac{gT}{L}$, so we can calculate the Thiamine volume that should be added to the medium after the malfunction:

$$V_T = \frac{W_T}{C_{\text{T stock}}} = \frac{105gT}{90 \frac{gT}{L}}$$

$$\boxed{\boldsymbol{V_T = 1.17\ L_T}}$$

<u>Question 3</u>

A factory carries out batch fermentation of bacteria for the purpose of biomass production in a 500 liter fermenter. Each fermentation cycle takes 24 hr and includes: 6 hr of fermenter cleaning and filling for fermentation, 1 hr sterilization of the fermenter and its content, 1 hr of lag phase, 15 hr of logarithmic growth and 1 hr of fermenter discharge. The doubling time of the bacteria in this fermentation is 50.3 min and the growth yield on oxygen is $1.2 \frac{gr\ Bacteria}{gr\ Oxygen}$.

The air supplied to the system runs through continuous high temperature sterilizer (K_d=17.39 min^{-1}) on its way into the fermenter, while the staying time of the air in the continuous sterilizer is 88 sec. The initial concentration of contaminating cells in the air (before sterilization) is 10^6 cells per 1 m^3 air, and the contamination risk level that the factory is willing to accept (after air sterilization) is 1 contamination per 1,000 fermentations.

During the fermentation, the oxygen percentage in the air which exits the fermenter is being continuously measured, and at the end of the fermentation the oxygen reading was 3.46%. It can be assumed that the air acts as ideal gas, in temperature of 25°C and pressure of 1 atmosphere.

A. What is the bacteria concentration at the end of the fermentation? *(5 gr/lit)*

B. If the cross section area (A) of the heat exchanger, where the air sterilization takes place, is 73.35 cm^2, what is the length of this heat exchanger (in meters)? *(25 m)*

<u>Solution</u>:

Given:

Air	**Bacteria**	**Fermenter**
$O_{2\,out_f} = 0.0346\dfrac{L_{O_2}}{L_{air}}$	$t_d = 50.3\text{min} = 0.84\text{hr}$	$V_{ferm} = 500L$
$\text{T} = 25°\text{C} = 298°\text{K}$	$Y_{\frac{X}{O}} = 1.2\dfrac{gX}{g_{O_2}}$	$t_T = 24hr$
$\text{P} = 1\text{atm}$		$t_{clean} = 6hr$
$N_0' = 10^6\dfrac{cells}{m_{air}^3}$		$t_{st} = 1hr$
$N_f = \dfrac{1}{1000}\dfrac{cells}{Ferm}$		$t_{lag} = 1hr$
$k_d = 17.39\dfrac{1}{min}$		$t_{log} = 15hr$
$t_{st} = 88\text{sec} = 1.47\text{min}$		$t_{harvest} = 1hr$

A. X_f =?

We can calculate X_f from the oxygen uptake at the end of the fermentation:

$$OUR_f = \frac{\mu \times X_f}{Y_{\frac{X}{O}}}$$

$Y_{\frac{X}{O}}$ is given in the question and μ can be calculated from the doubling time:

$$\mu = \frac{ln2}{t_d}$$

$$\mu = \frac{ln2}{0.84\text{hr}} = 0.825\frac{1}{hr}$$

We need the oxygen uptake at the end of the fermentation, which can be calculated from the change in the oxygen percentage in the air exiting the fermenter:

$$OUR = \frac{Q_{O_2}}{V}_{in} - \frac{Q_{O_2}}{V}_{out}$$

We can calculate the air flow rate into the fermenter from the sterilization data and the fermentation requirements.

From the sterilization equation we can find the number of contaminating cells entering the sterilizer:

$$ln\frac{N_1}{N_0} = -k_d \times t_{st}$$

$$ln\frac{\frac{1}{1000}\frac{cells}{Ferm}}{N_0} = -17.39\frac{1}{min} \times 1.47\text{min} \to N_0 = 1.2 \times 10^8\frac{cells}{Ferm}$$

We can calculate the air volume supplied to the fermenter by using the contaminating cells concentration in the air:

$$V_{air} = \frac{N_0}{N_0'}$$

$$V_{air} = \frac{1.2 \times 10^8\frac{cells}{Ferm}}{10^6\frac{cells}{m_{air}^3}} = 120\frac{m_{air}^3}{Ferm}$$

The air is supplied into the fermenter during the growth stages, therefore the air flow rate:

$$Q_{air} = \frac{V_{air}}{t_{lag} + t_{log}}$$

$$Q_{air} = \frac{120\frac{m_{air}^3}{Ferm}}{(1+15)hr} = 7.5\frac{m_{air}^3}{hr} \times 1000\frac{L_{air}}{m_{air}^3} = 7{,}500\frac{L_{air}}{hr}$$

The question does not detail the oxygen percentage in the supplied air so we will assume it is normal air with 21% oxygen, hence the oxygen uptake rate at the end of the fermentation:

$$OUR_f = \frac{Q_{O_2}}{V_{in}} - \frac{Q_{O_2}}{V_{out_f}} = \frac{Q_{air}}{V_{ferm}} \cdot \frac{P}{R \cdot T} \cdot Mw_{O_2} \cdot \left(\%O_{2_{in}} - \%O_{2_{out_f}}\right)$$

$$OUR_f = \frac{7{,}500\,\frac{L_{air}}{hr}}{500L} \cdot \frac{1atm}{298°K \cdot 0.082\,\frac{L_{O_2} \cdot atm}{°K \cdot mole_{O_2}}} \cdot 32\,\frac{g_{O_2}}{mole_{O_2}} \cdot (0.21 - 0.0346)\,\frac{L_{O_2}}{L_{air}} = 3.44\,\frac{g_{O_2}}{L \cdot hr}$$

Therefore, the biomass concentration at the end of the fermentation:

$$OUR_f = \frac{\mu \times X_f}{Y_{\frac{X}{O}}} \rightarrow X_f = \frac{OUR_f \times Y_{\frac{X}{O}}}{\mu}$$

$$X_f = \frac{3.44\,\frac{g_{O_2}}{L \times hr} \times 1.2\,\frac{gX}{g_{O_2}}}{0.825\,\frac{1}{hr}}$$

$$\boxed{\boldsymbol{X_f = 5\,\frac{gX}{L}}}$$

B. $\boldsymbol{l} =?$

Given that $A = 73.35cm^2$

The length of the heat exchanger we can find from the staying time in the fermenter:

$$t_{st} = \frac{l}{U}$$

While t_{st} is given in the question and represents the U air flow velocity.

The air flow velocity we can calculate from the flow rate that was calculated in section A:

$$U = \frac{Q}{A}$$

$$U = \frac{7.5\,\frac{m^3_{air}}{hr}}{73.35cm^2} \times 10^4\,\frac{cm^2}{m^2} = 1022.5\,\frac{m}{hr}$$

$$U = \frac{1022.5\,\frac{m}{hr}}{60\,\frac{min}{hr}} = 17\,\frac{m}{min}$$

Therefore, the length of the heat exchanger is:

$$l = t_{st} \times U$$

$$l = 1.47min \times 17 \frac{m}{min}$$

$$\boxed{\boldsymbol{l = 24.99\ m \cong 25\ m}}$$